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Resolving Structural Imaging Distortions With Elastic Full-Waveform Inversion in Middle East Ultra-Shallow Water Environments

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Abstract

Structural imaging distortions are common in regions with significant velocity heterogeneity and complex geological features, such as karsts and channels in the overburden, as observed offshore Abu Dhabi. These regional challenges have hindered full-waveform inversion (FWI) from accurately resolving small-scale structural distortions, resulting in increased uncertainties in geological interpretations of target reservoirs. The ultra-shallow-water environments and the hard seabed introduce significant mud-roll, guided waves, and other elastic effects in the recorded data. These effects challenge conventional FWI based on acoustic approximations, often requiring dedicated workflows that incorporate more robust cost functions. However, accurately modeling shallow heterogeneities and strong contrasts using an elastic engine is expected to improve inversion results. Additionally, the lack of diving-wave penetration due to significant near-surface velocity contrasts often confines diving-wave-driven FWI updates to the very shallow. The lack of low-frequency signals in these settings further complicates the inversion process.

This paper proposes an FWI workflow that integrates surface-wave inversion (SWI) to develop a reliable near-surface starting model, while leveraging both reflections and diving waves with a time-lag cost function. The workflow is demonstrated on the mega high-density ocean bottom node (OBN) dataset from offshore Abu Dhabi, where it enhances inversion stability and improves FWI convergence for deeper sections. Our results show that this OBN data-driven workflow, even at 12 Hz with an acoustic engine, significantly improves the continuity and focusing of conventionally migrated images, effectively resolving structural distortions caused by complex karst formations and deeper channels.

We present a 30 Hz acoustic FWI result that further mitigates many of the deeper distortions observed in conventional imaging, caused by shallow complexities. Furthermore, we demonstrate that elastic FWI further enhances the conventional image, particularly in the areas with strong velocity contrasts, where elastic effects are most pronounced. Finally, we compare the 30 Hz FWI results from both acoustic and elastic approaches, showing that both outperform conventional imaging, with the elastic result having the best event continuity and signal-to-noise ratio from shallow to deep.

Introduction

Since its introduction in the 1980s, full-waveform inversion (FWI) has been a key technique for recovering subsurface parameters by minimizing the misfit between modeled and recorded seismic data. Over the years, FWI has evolved significantly, driven by advances in both computational power and technology, establishing itself as a cornerstone for velocity model building in the industry. This evolution has led to the development of alternative cost functions, incorporation of additional subsurface complexities (such as viscosity), and a push for high-frequency model updates (e.g., [Warner and Guasch, 2014](#); [Metivier et al., 2016](#); [Zhang et al., 2018](#); [Espin et al., 2023](#)). Notably, the recent advancements in elastic FWI have positioned it as a state-of-the-art technique for constructing highly detailed and accurate velocity models. In contrast to the more commonly used acoustic FWI, elastic FWI has demonstrated superior performance in regions with significant velocity contrasts (e.g., [Plessix and Krupovnickas, 2021](#); [Wu et al., 2022](#)).

Ocean-bottom node (OBN) surveys capture both compressional and converted-wave events, and the elastic effects introduce phase and amplitude changes to the wavefield. Elastic FWI enables the incorporation of wavefield complexities, such as mode conversions (P-to-S and potentially back to P), into the inversion process. In acoustic FWI, these conversions are typically not modeled, which can lead to misfitting of data and cycle-skipping issues. As a result, elastic FWI has the potential to provide cleaner velocity updates. For example, [Solano and Plessix \(2023\)](#) demonstrated the effectiveness of OBN data in elastic FWI.

The successful application of FWI in marine environments, even in geologically complex regions, has been well-documented (e.g., [Wang et al., 2019](#); [Chen et al., 2024](#)). However, offshore Abu Dhabi presents unique challenges for FWI. These include a hard, rugose, ultra-shallow water bottom, and a heterogeneous near-surface that generates strong interference from mud-roll and guided waves ([Figure 1a](#)). Additionally, pronounced velocity contrasts in the shallow subsurface—due to thinly laminated sequences of carbonates, anhydrite, and shale layers—limit the penetration of diving waves ([Figure 1b](#)) and introduce shingling effect in the recorded data, with the true first arrivals disappearing at the far offset due to a velocity reversal in the near surface. Structural distortions, such as karst formations and channels, further hinder imaging of deeper targets. Other challenges include the absence of low-frequency signals due to size limitations of seismic sources in shallow water and the inherent poor signal-to-noise ratio in such environments. These complexities often destabilize FWI, making inversion convergence difficult. While recent studies (e.g., [Hermant et al., 2022](#); [Damianus et al., 2023](#)) have reported successful FWI applications in offshore Abu Dhabi, small-scale structural distortions have often been poorly resolved. [Hutahayan et al. \(2023\)](#) demonstrated that with sufficient noise attenuation, shallow inversion stability can be achieved, but diving-wave-based FWI remains robust only for the upper 500 meters due to limited penetration and shingling effects. This highlights the need for a high-resolution FWI model that can correct small-scale kinematic errors beyond the diving-wave penetration depth.

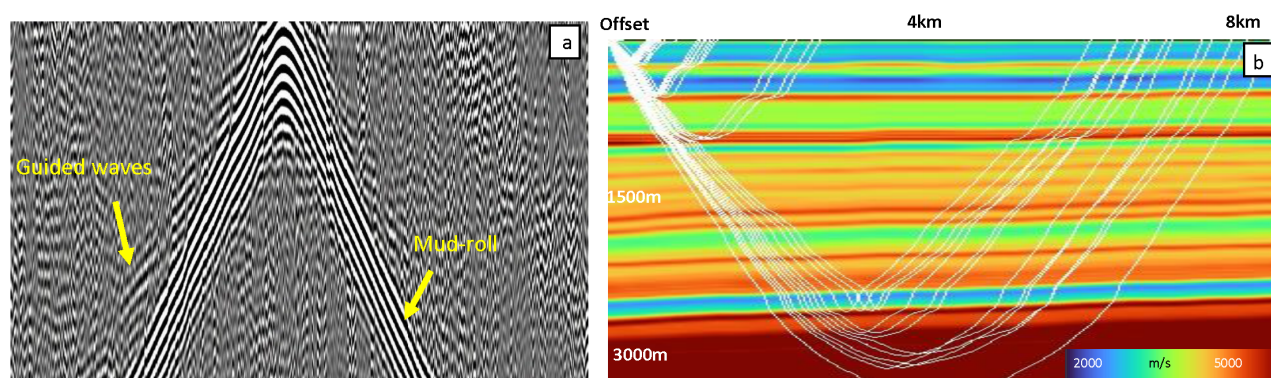


Figure 1—Examples of data challenges in offshore Abu Dhabi: (a) raw pressure data with 6 Hz low-pass filter, and (b) diving-wave ray tracing through a smoothed model extrapolated from a well.

In this paper, we compare acoustic and elastic FWI applied up to 30 Hz using OBN data from Abu Dhabi's ultra-shallow water environments. We propose a workflow that simultaneously inverts for diving waves and reflection energy to capture deeper velocity variations and enhance model resolution. In this study, we apply time-lag FWI (TLFWI), which uses a time-lag cost function to mitigate issues such as amplitude mismatch, cycle skipping, and the challenges associated with limited low-frequency content (Zhang et al., 2018). In cases where low-frequency signals are absent and diving-wave coverage is limited, constructing a robust starting model becomes critical. However, creating a reliable near-surface velocity model in such geologically complex settings is difficult. Refraction-based tomography often struggles with instability due to shingling effects, while reflection-based tomography lacks sufficient near-surface kinematic data due to inadequate pre-critical reflection energy. To address these issues, we employed surface-wave inversion (SWI) to improve near-surface velocity resolution, followed by long-wavelength tomography to derive the FWI starting model.

Data challenges

The data acquisition took place in 2023 offshore Abu Dhabi, using a receiver grid of 25 m $\times$ 150 m, a shot grid of 25 m $\times$ 25 m, and a full-azimuth, full-fold coverage over an area of 1653 km², achieving a nominal fold of 9600. The maximum offset in both the x and y directions is 6 km, with selected receivers reaching up to 16 km. At the time of writing, this represents the largest and densest OBN acquisition. The water depths range from 4 m to 33 m, with an exceedingly large reflection coefficient of approximately 0.6. Due to the ultra-shallow water environment, the use of larger seismic guns is restricted, resulting in the deployment of 1520 cu.in. array for this survey, which limits the low-frequency content in the recorded data. The environment also generates prominent mud-roll and guided waves, which interfere with the diving waves originating from deeper sections. This interference is laterally highly variable, due to the horizontal heterogeneity of the near-surface geology. The large velocity contrasts across all the depths, particularly at the seabed and in the near surface, greatly limits the diving waves, as well as being the source of substantial converted waves, leading to elastic complications. The data underwent comprehensive conditioning, including linear noise attenuation and random noise suppression. Despite these efforts, FWI based on diving waves alone struggled below 500 m, as the diving waves could not be fully separated from the complex wave interference originating from the near surface. To address this, reflection energy was incorporated into the inversion process to better capture deeper velocity variations.

Starting model and use of SWI

Incorporating an accurate near-surface model into the starting model, as demonstrated in previous studies, could enhance deeper imaging and improve the convergence of FWI by reducing cycle-skipping challenges

(Guo and Aziz, 2024). To achieve this, we apply SWI, which utilizes mud roll dispersion curves to derive a near surface V_s model (Li et al., 2019). This V_s model is then converted to a near-surface V_p model using a spatially variant V_p -to- V_s relationship extracted from well information. To finalize the starting model, long-wavelength tomography was applied after incorporating the near-surface V_p model derived from SWI. Anisotropy parameters were derived from well data using Backus averaging (Liner, 2014) and refined through anisotropy scans. As shown in Figure 2, the inclusion of SWI in the starting model reduces potential cycle-skipping issues by enhancing the continuity of the model, as indicated by the arrows.

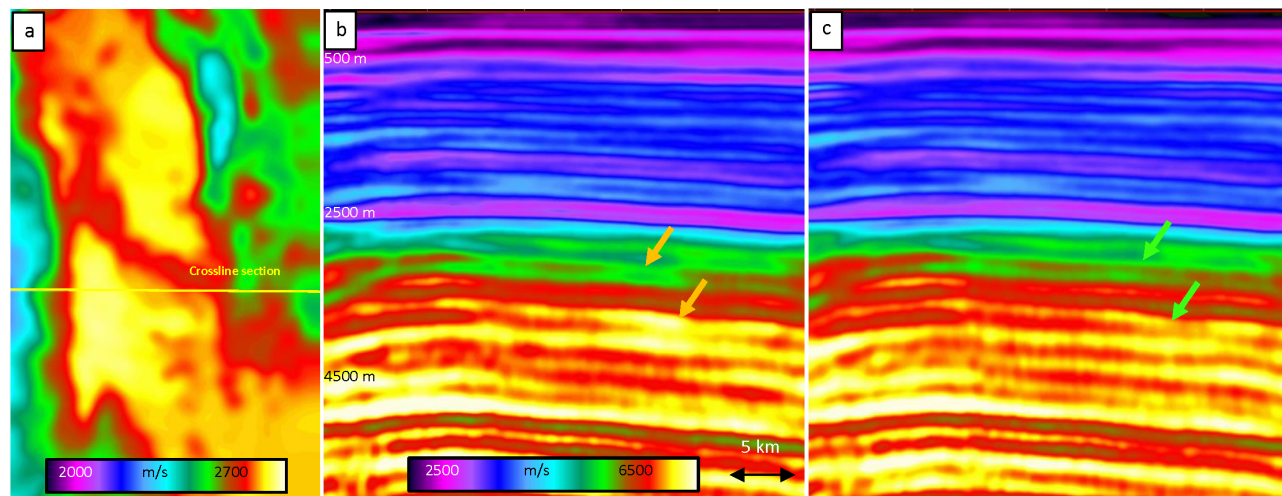


Figure 2—(a) Depth slice at 60 m for starting model with SWI incorporated in the shallow. Comparison of resulting 12 Hz FWI models: (b) without SWI in the starting model, and (c) with SWI in the starting model.

Enhancing FWI convergence with reflections and time-lag cost function

FWI using reflections presents notable challenges, as reflection-based updates often lack essential low-wavenumber components needed for accurate kinematic corrections. The amplitude discrepancies between recorded and synthetic data are particularly pronounced for reflections, especially when dealing with reflectors with strong contrast. To address this, TLFWI, which emphasizes the kinematics between modeled and recorded data, was implemented. Zhang et al. (2018) demonstrated that TLFWI significantly enhances FWI with reflections, leading to reduced cycle-skipping and improved convergence. Figure 3 compares the results of 10 Hz TLFWI using diving-waves only versus using both diving waves and reflections. The starting model (Figure 3a) is a smoothed legacy model. Both FWI approaches improved image focusing and reduced long-wavelength undulations in the reverse-time migration (RTM) QCs shown in Figures 3e and 3f. While the near-surface results between the diving-wave-only FWI (Figure 3b) and the full-wavefield FWI (Figure 3c) are quite similar, the full-wavefield FWI exhibited better horizontal and vertical resolution. Figures 3e and 3f highlight the imaging improvements from including reflections, showing better resolution of distortions caused by the slow karst velocity and undulations in the circled target area.

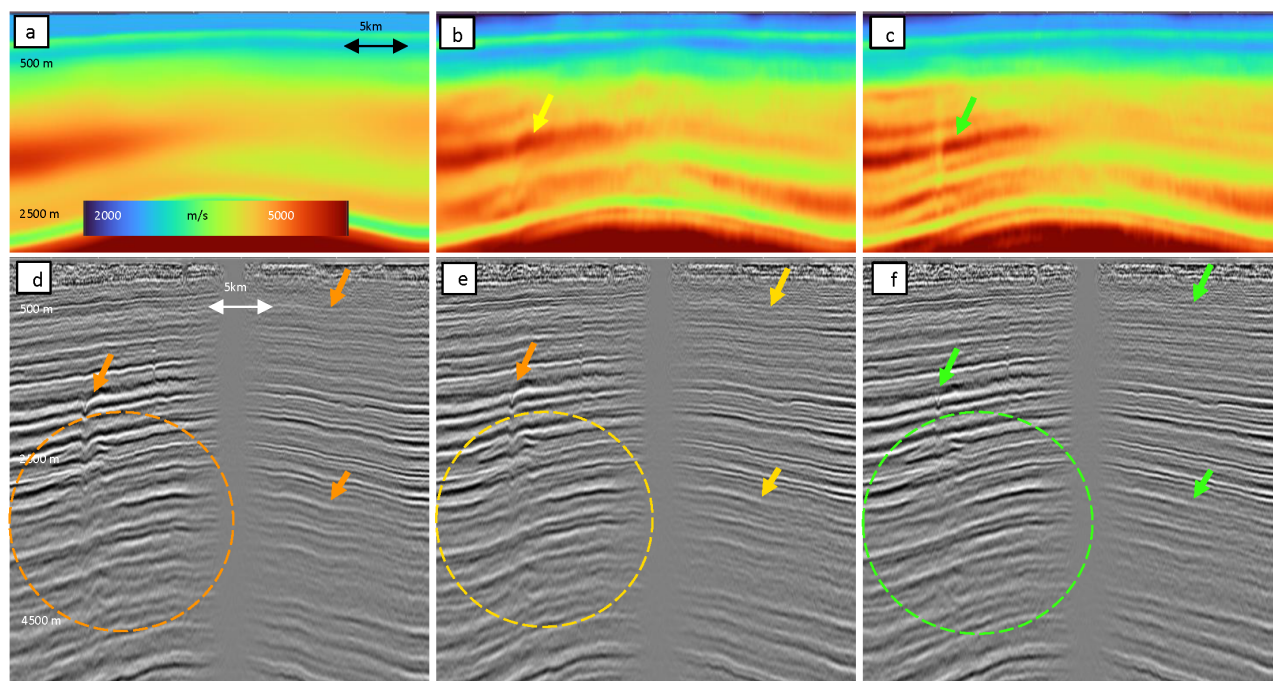


Figure 3—Comparison of 10 Hz FWI models: (a) starting model, (b) FWI with diving waves only, (c) FWI with diving waves and reflections. RTM stacks migrated with: (d) starting model, (e) FWI with diving waves only, and (f) FWI with diving waves and reflections.

12 Hz FWI - acoustic

The 12 Hz acoustic FWI workflow integrates SWI for the near surface and utilizes both diving waves and reflections with a time-lag cost function. Despite the challenges mentioned earlier, the inversion achieved markedly high horizontal and vertical resolution (Figure 4). The workflow proved to be stable even at depths of ~6-7 km, improving model resolution while effectively correcting structural distortions and undulations caused by velocity variations across the area. FWI successfully addressed major distortions beneath the slow karst layers, which are key drilling hazards. The Kirchhoff migration comparison in Figures 4d and 4e showed substantial improvements, including better focusing, simplified geological structures, and more continuous events. These improvements were particularly evident in the deeper target section of the anticline, where geological conformability was significantly enhanced. This FWI model not only added substantial value for interpretation and reservoir characterization but also provided essential insights for mitigating drilling risks.

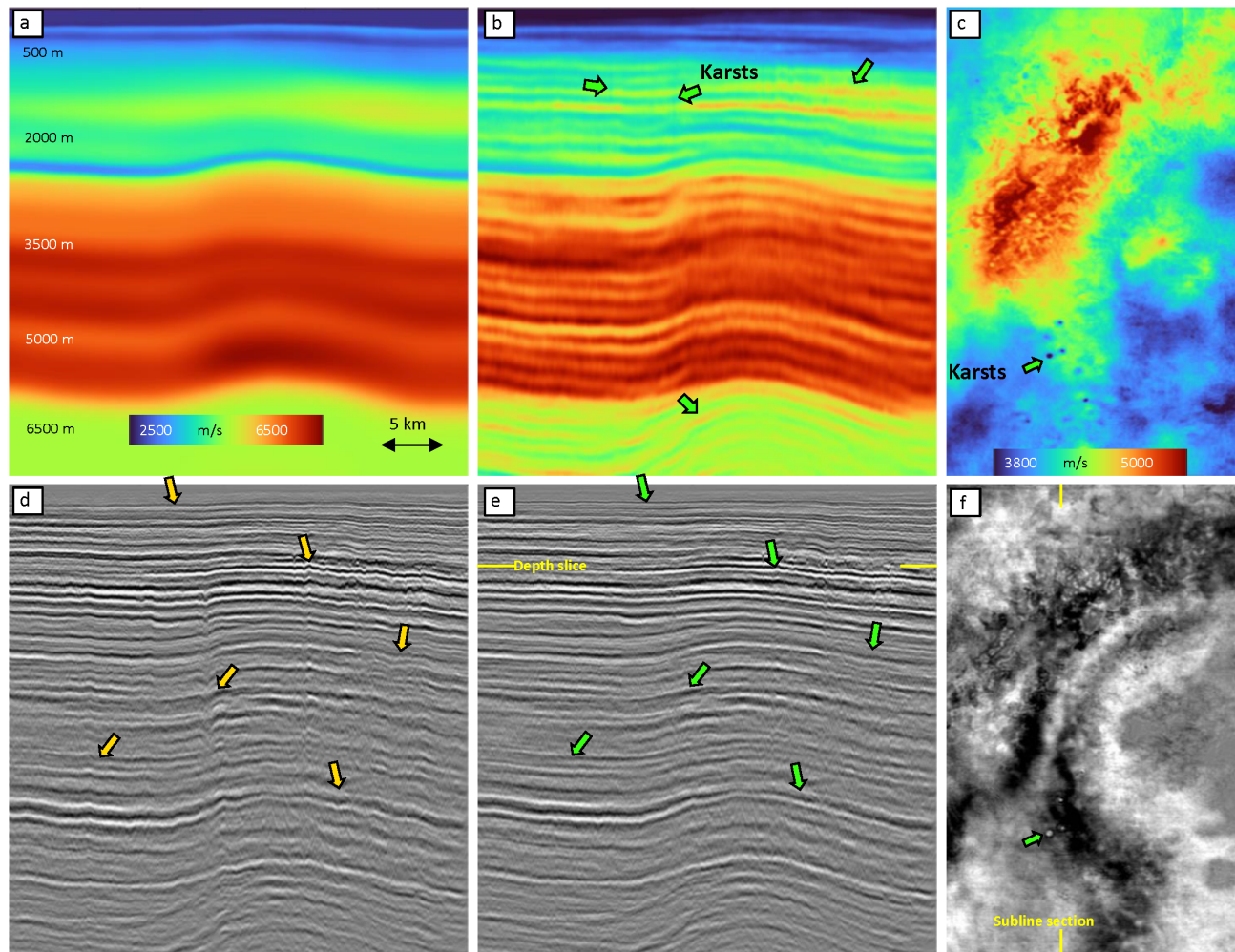


Figure 4—12 Hz FWI model: (a) starting, and (b) FWI model. Kirchhoff depth stacks migrated with: (d) starting, and (e) FWI model. Depth slices at 1200 m: (c) FWI model, and (f) migrated stack with FWI model.

30 Hz FWI - acoustic and elastic

The 30 Hz V_p models from acoustic and elastic TLFWI are compared in Figure 5. The 30 Hz FWI captures high frequency components of the velocity model, correcting distortions and undulations beneath small-scale velocity anomalies, as demonstrated by Figures 5b and 5e. However, in the most kinematically complex area, highlighted by the leftmost and rightmost arrows, under the clinoform wedges and channels respectively, some imaging distortions and undulations remain. These heterogeneous geobodies are strong generators of elastic effects and acoustic FWI alone cannot resolve the velocity anomalies fully. As a result, elastic FWI was performed.

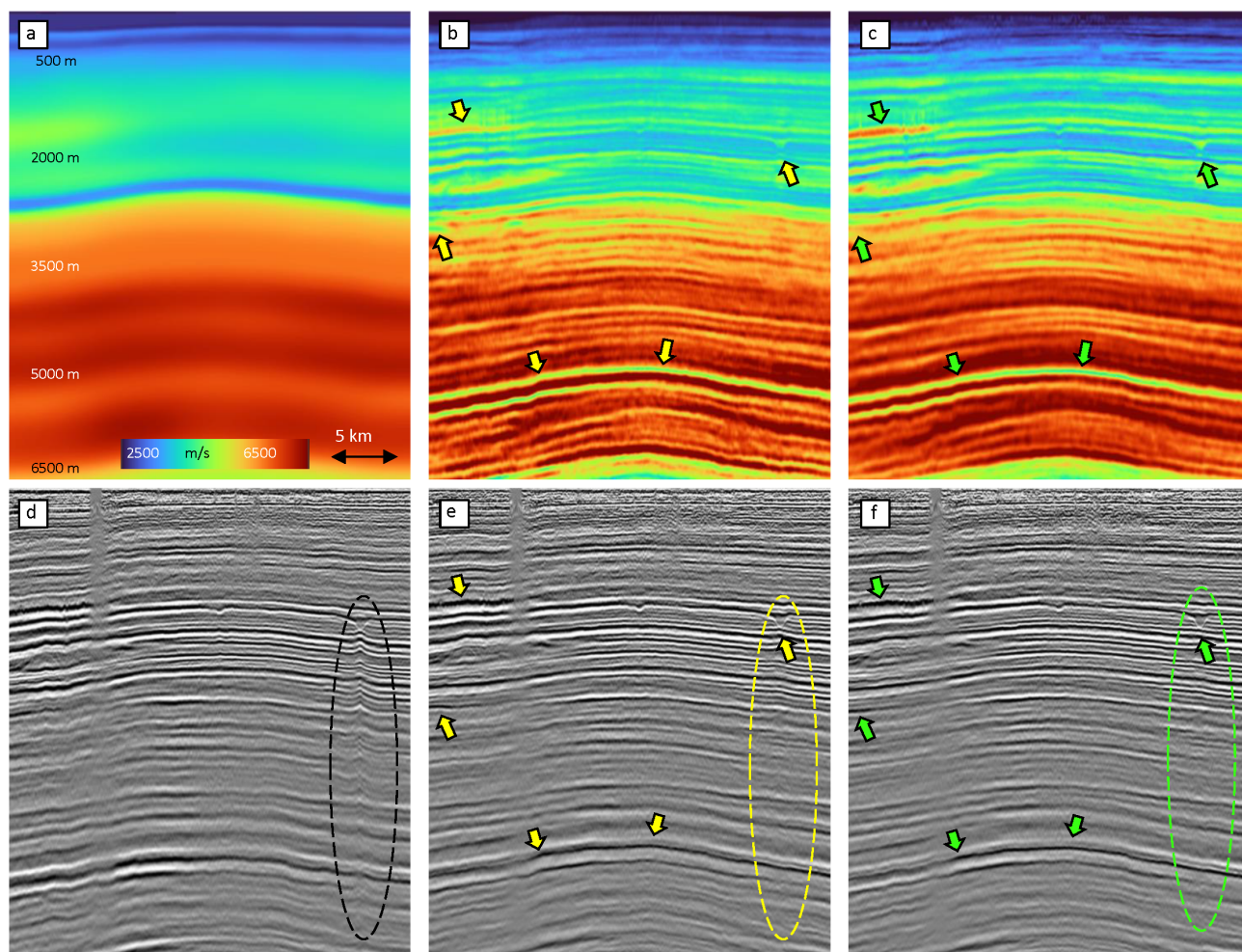


Figure 5—Comparison of 30 Hz FWI models: (a) starting model, (b) acoustic FWI, and (c) elastic FWI. 40 Hz RTM stack migrated with: (d) starting model, (e) acoustic FWI, and (f) elastic FWI.

For the elastic inversion, a spatially varying V_p/V_s ratio is used based on an empirical relationship derived from wells. To further improve the modelling, the V_s model is passively updated throughout the inversion to preserve this V_p/V_s ratio. With the proposed workflow, most of the structural distortions on the RTM image can be resolved using 30 Hz acoustic FWI alone (Figures 5d and 5e). In both cases, TLFWI has produced geologically conformable models of unprecedentedly high resolution for this region. Elastic FWI further improves the results by producing a cleaner model, with more geological continuity, enhancing the definition of strong contrasts such as channels, and showing greater stability near complex features like karsts and wedges, as indicated by the two arrows on the left of Figures 5c and 5f. The sharp velocity contrasts in the deeper section, highlighted by the bottom two arrows in Figures 5e and 5f, are effectively resolved by elastic FWI, resulting in improved kinematic accuracy in the FWI model and a simplified structural image.

FWI Image

At this frequency, an FWI Image (Zhang et al., 2020) can be generated from the velocity models to complement the conventional seismic image. Figures 6a and 6b show that the elastic FWI Image exhibits improved resolution and a higher signal-to-noise ratio compared to the acoustic FWI Image, highlighting its ability to more accurately capture detailed subsurface features. In the shallow section, the elastic FWI Image is comparable to the RTM image in most areas while having better signal to noise ratio; the differences become more apparent in regions marked by cyan arrows and boxes. These zones suffer from poor illumination caused by acquisition gaps and complex geological structures, where conventional imaging

struggles. The FWI Imaging approach improves the imaging quality here by fully exploiting the complete wavefield, including diving waves, reflections, and converted waves, as shown in Figures 6b and 6c.

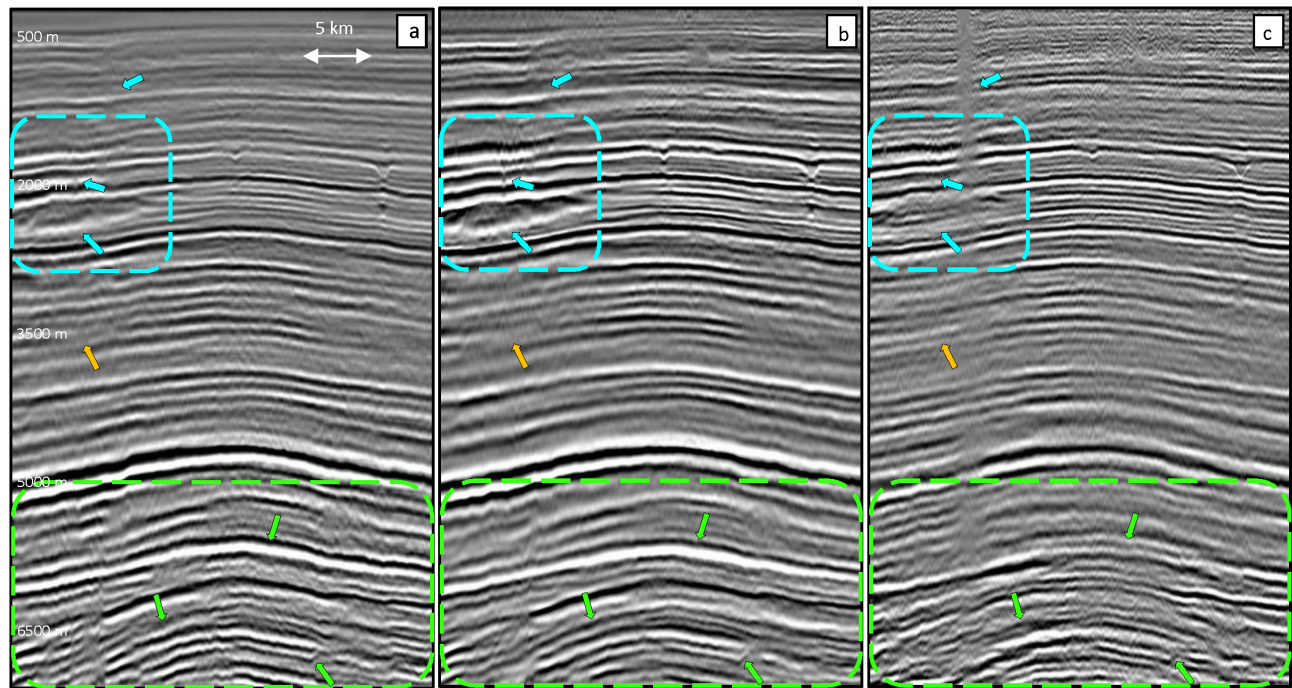


Figure 6—Comparison of 30 Hz subsurface images: (a) acoustic FWI Image, (b) elastic FWI Image, and (c) RTM image migration with the 30 Hz elastic FWI model.

In the deeper section, particularly in the green-highlighted area beneath strong reflectors, the 30 Hz elastic FWI Image displays superior imaging relative to the 30 Hz RTM image because the RTM in this zone is degraded by interbed multiples. These interbeds are more effectively handled by the inversion process in FWI, leading to a cleaner and more accurate representation of the subsurface. Nonetheless, some challenges remain beneath the most complex geological bodies. As indicated by the orange arrows in Figure 6, the FWI Image displays somewhat reduced amplitude continuity compared to the RTM image, suggesting that further refinement of the inversion parameters or advanced modeling techniques such as elastic FWI for Vs update could yield additional improvements. Overall, these observations underline the complementary nature of FWI Imaging, especially with elastic FWI, in resolving complex subsurface features that conventional imaging methods alone may not fully capture.

Conclusions

This study adopts an elastic FWI workflow that incorporates SWI to address near-surface velocity complexities and applies TLFWI to improve the utilization of reflection energy. As a result, even in complex and heterogeneous ultra-shallow-water environments, the method produces a clean, high-resolution velocity model that resolves structural distortions and enhances the continuity and focusing of subsurface images. The results demonstrate that most kinematic errors are corrected at the lower frequency of 12 Hz. However, inverting velocity up to 30 Hz further reduces kinematic errors by capturing higher frequency components of the velocity model. Compared to acoustic FWI, elastic FWI improves event continuity, resolution, and the signal-to-noise ratio. The elastic FWI model and the resulting FWI Images provide significant value for interpretation and reservoir characterization, while also offering essential insights for mitigating drilling hazards. Future work will focus on exploring a dedicated multi-component Vs inversion using elastic FWI to further improve the Vp update.

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